# DDoS Attack Detection Using ResNeXt50-32x4d with MindSpore


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***Abstract*—Distributed Denial of Service (DDoS) attacks exploiting the TCP SYN handshake remain a persistent threat to network availability. This paper presents *SYNSheild*: an end-to-end deep learning pipeline that converts raw network packet captures (PCAP) into fixed-size $224 \times 224$ grayscale images and classifies them as DDoS or benign using a ResNeXt-50 (32×4d) convolutional neural network trained with Huawei MindSpore and the *mindcv* toolkit. The primary model is trained on 0.3 s time-window packet images derived from the public CICDDoS2019 benchmark. On a held-out test set of 1845 samples it achieves 98.10% accuracy and a macro F1 score of 0.9375, with 100% recall on the DDoS class. A blind random sample of 270 balanced test images (seed 42) yields 98.9% accuracy and macro F1 0.9889. Cross-domain experiments confirm that train/deploy representation alignment is mandatory; key findings and operational caveats are documented alongside reproducible evaluation scripts.**



**Keywords: SYN flood; DDoS detection; packet-to-image; ResNeXt-50; MindSpore; CICDDoS2019; intrusion detection; deep learning.**


## I. Introduction

TCP SYN flood attacks exploit the three-way handshake to exhaust server connection-state resources by sending large volumes of SYN segments without completing the handshake. Despite their relative simplicity, SYN floods account for a substantial fraction of volumetric DDoS incidents observed in operational networks [1]. Signature-based and shallow machine-learning intrusion detection systems (IDS) struggle with obfuscated or novel attack variants, making automated, data-driven detection an active research challenge.

A compelling direction is the *packet-to-image* paradigm: raw packet bytes are arranged as two-dimensional pixel matrices and fed into a convolutional neural network (CNN), allowing the model to learn discriminative spatial structure without manual feature engineering [2], [3]. Building on this foundation, this paper replaces the ResNet-50 architecture used in earlier BAU work [2] with **ResNeXt-50 (32×4d)**—a grouped-convolution variant offering richer representational capacity at similar parameter counts [4]—and implements the entire training and inference stack on **Huawei MindSpore** [5] with *mindcv* [6].

**Contributions.**

- A reproducible PCAP-to-checkpoint pipeline from CICDDoS2019 zips to trained model, including SYN-based heuristic labeling, packet-to-image conversion, and class-imbalance-aware training.
- A **0.3 s time-window** image encoding as the primary training representation, with focal-loss [7] training and DDoS-F1-oriented early stopping.
- Quantitative evaluation including full split metrics, blind inference, calibration diagnostics, and a cross-domain mismatch study.
- First open, reproducible DDoS image classifier built entirely on MindSpore, supporting deployment on CPU and GPU (with an Ascend-compatible graph export path).

The rest of the paper is organized as follows. Section II surveys related work. Section III describes the system design and methodology. Section IV presents experimental results. Section V discusses limitations and operational caveats. Section VI concludes with future directions.

## II. Related Work

### A. DDoS Detection: Classical and Feature-Based Methods

Early DDoS detection relied on rule-based thresholds and statistical features such as SYN/ACK ratios or packet rates [1]. While fast and interpretable, these approaches are brittle against rate-varied or low-and-slow attacks. Machine learning (ML) on engineered flow features improves adaptability but requires substantial domain expertise for feature selection and is sensitive to feature drift.

### B. Deep Learning for Network Intrusion Detection

Deep learning methods have been applied broadly to intrusion detection, spanning recurrent models for sequential traffic, autoencoders for anomaly scoring (e.g., Kitsune [8]), and hybrid CNN–RNN architectures reviewed in [9]. SDN-integrated deep learning detectors [10] demonstrate near-real-time capability but depend on controller telemetry not always available in classical networks.

### C. Packet-to-Image Representations

Lotfollahi et al. [3] showed that raw byte sequences fed to CNNs can classify encrypted traffic without decryption. Bazzi et al. [2] applied this idea specifically to SYN flood detection on CICDDoS2019 [11], encoding individual packet bytes as 224×224 grayscale images and classifying with ResNet-50 [12], achieving strong benchmark accuracy. Our work extends this line by (i) upgrading to ResNeXt-50 [4], (ii) using *time-window* aggregation for temporal context, (iii) deploying on MindSpore, and (iv) systematically studying cross-domain representation transfer.

### D. Summary Comparison

Table I positions this work relative to representative prior approaches.

TABLE I
QUALITATIVE COMPARISON WITH REPRESENTATIVE DETECTION APPROACHES.

| Method | Strengths | Limitations |
|---|---|---|
| Rules / thresholds | Fast, explainable | Brittle to novel attacks |
| ML on flow features | Data-adaptive | Feature engineering burden |
| Kitsune [8] | Unsupervised anomaly | No DDoS-specific supervision |
| ResNet packet-image [2] | Raw-byte patterns; strong CIC results | Single architecture; per-packet only |
| **This work** | ResNeXt; time-window; MindSpore; full pipeline | Lab dataset; same-domain eval |

## III. SYSTEM DESIGN AND METHODOLOGY

### A. Dataset

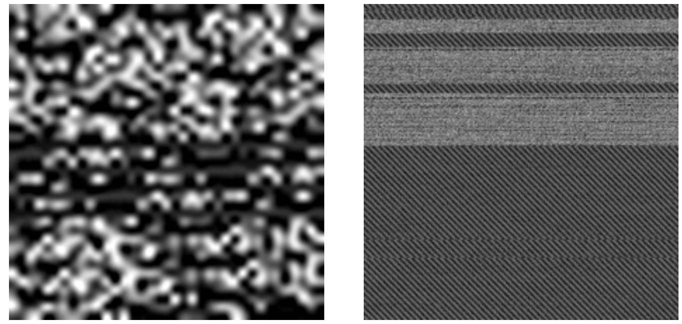
We use the **CICDDoS2019** dataset [11] from the Canadian Institute for Cybersecurity, which contains realistic benign and attack traffic including multiple SYN flood scenarios captured under controlled laboratory conditions. Raw data are provided as PCAP archives and processed through the pipeline described below.

### B. PCAP Labeling

A lightweight SYN-count heuristic labels each PCAP before image conversion: any PCAP with more than **100 TCP SYN packets** is assigned the *DDoS* class; otherwise it is labeled *normal*. This threshold was chosen to reflect the high-volume nature of SYN floods and to avoid false positives from connection-heavy but benign services. The labeling script (`classify_pcap_syn.py`) performs an early exit once the threshold is exceeded, making it fast even on large captures.

### C. Packet-to-Image Encoding

For DDoS PCAPs, only TCP SYN packets are extracted (`–syn-only`) to isolate the attack signature; for normal PCAPs all packets are used. Raw packet bytes are arranged sequentially into a 2D array and resized to $\mathbf{224 \times 224}$ grayscale pixels using Pillow, following the methodology in [2]. During training, grayscale images are replicated to three channels to match the ImageNet-pretrained normalization expected by the backbone. Figure 1 illustrates example images from each class.



(a) Normal (benign) traffic

(b) SYN-flood DDoS traffic

Fig. 1. Representative $224 \times 224$ grayscale packet images from the primary 0.3 s time-window pipeline on CICDDoS2019.

### D. Time-Window Aggregation

The **primary model** uses a **0.3 s time-window** scheme: packets captured within a rolling 0.3 s interval are aggregated into a single image, providing temporal context across multiple packets within the window. The resulting image set is stored under `dataset/images_per_second_window_0p3/` with `train/`, `val/`, and `test/` subdirectories, each containing `ddos/` and `normal/` class folders. A legacy **per-packet** configuration (one image per individual packet) is retained for comparison.

### E. Model Architecture

The classifier is **ResNeXt-50 (32×4d)** [4], instantiated via *mindcv* [6] with a two-unit classification head (DDoS vs. normal) and a softmax output layer. ResNeXt extends ResNet by replacing standard $3 \times 3$ convolutions with *grouped* convolutions parameterized by a cardinality $C = 32$ and per-group width $d = 4$, which increases representational diversity without proportionally increasing model size [4].

### F. End-to-End Pipeline

Figure 2 summarizes the complete flow from raw captures to inference decision.

### G. Training Configuration

Training is implemented in `notebook/train_resnext.py` and launched by `scripts/train_resnext_per_second_0p3.sh`. Table II details the primary hyperparameter choices.

**Class imbalance.** The CICDDoS2019 0.3 s split is heavily skewed toward normal traffic (1710 normal vs. 133 DDoS in validation). We address this with **focal loss** combined with an **8× minority boost** on the DDoS class, and use DDoS-class F1 on the validation set as the early-stopping criterion to prioritize attack recall.

## IV. EXPERIMENTAL RESULTS

### A. Experimental Setup

All offline evaluations were recorded on **2026-04-03** using the `focal_v2` primary checkpoint. Hardware: **NVIDIA**

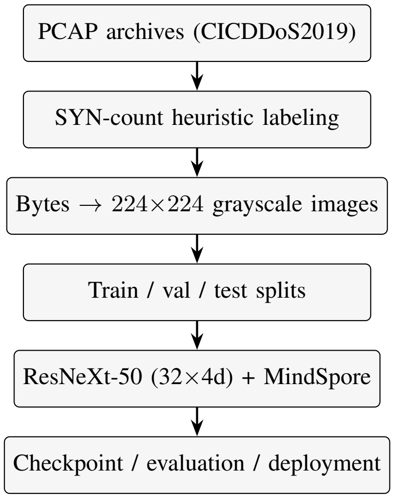



Fig. 2. End-to-end *SYNSheild* pipeline: raw network captures to inference decision via packet-image classification.

TABLE II
PRIMARY TRAINING HYPERPARAMETERS
(PER_SECOND_0P3_FOCAL_V2).

| **Hyperparameter** | **Value** |
|---|---|
| Framework | Huawei MindSpore 2.6.0 (Graph mode) |
| Backbone | ResNeXt-50 (32×4d) via mindcv |
| Loss function | Focal loss [7] ($\gamma$=2.0, minority boost 8×) |
| Early stopping | Val DDoS-F1 (`f1_ddos`), patience 10 epochs |
| Max epochs | 40 |
| Batch size | 32 |
| Optimizer | Adam (cosine LR schedule) |
| Learning rate | $5\times10^{-4}$ decaying to $10^{-6}$ (1 warm-up epoch) |
| Augmentation (train) | RandomResizedCrop(224), RandomHorizontalFlip |
| Preprocessing (eval) | Resize 256 → CenterCrop 224, ImageNet norm. |
| Checkpoint metric | Macro F1 on validation set |

**GeForce RTX 3070**; software: MindSpore 2.6.0, mindcv 0.3.0, Python 3.10.12, CUDA 11.6. The one-shot driver is `scripts/run_all_eval_0p3.sh`; fixed seed 42 ensures reproducibility.

### B. Primary Model: 0.3 s Time-Window

Table III presents overall accuracy, macro F1, weighted F1, and balanced accuracy on the validation and test splits.

TABLE III
OVERALL METRICS FOR THE PRIMARY 0.3 S WINDOW MODEL.

| **Split** | $N$ | **Acc.** | **Macro F1** | **Bal. Acc.** |
|---|---|---|---|---|
| Validation | 1842 | 98.21% | 0.9399 | 0.990 |
| Test | 1845 | 98.10% | 0.9375 | 0.990 |

### C. Confusion Matrices

Table IV provides the raw confusion counts for both splits. Notably, the model achieves **zero missed DDoS samples** (false negatives) on both validation and test, at the cost of 33 and 35 false positives, respectively.

TABLE IV
CONFUSION MATRICES (ROWS = TRUE CLASS, COLS = PREDICTED CLASS).

| | **Validation** | | **Test** | |
|---|---|---|---|---|
| **True\Pred** | DDoS | Normal | DDoS | Normal |
| DDoS | **133** | 0 | **135** | 0 |
| Normal | 33 | **1676** | 35 | **1675** |

### D. Per-Class Precision, Recall, and F1

Table V shows per-class metrics derived from the confusion matrices. The DDoS class achieves **recall = 1.000** (no missed attacks) in both splits, which is the operationally critical metric for a security detector.

TABLE V
PER-CLASS PRECISION, RECALL, AND F1 ON VALIDATION AND TEST.

| | **Validation** | | | **Test** | | |
|---|---|---|---|---|---|---|
| **Class** | Prec. | Rec. | F1 | Prec. | Rec. | F1 |
| DDoS | 0.801 | 1.000 | 0.890 | 0.794 | 1.000 | 0.885 |
| Normal | 1.000 | 0.981 | 0.990 | 1.000 | 0.980 | 0.990 |
| Macro | | 0.940 | | | 0.937 | |

Figure 3 visualizes the per-class precision, recall, and F1 on the test split.

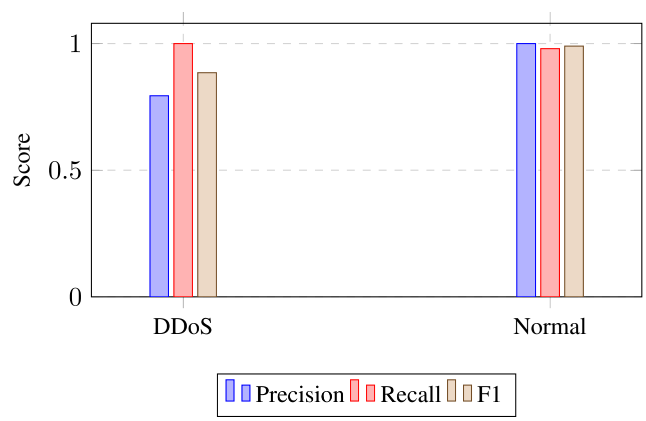



Fig. 3. Per-class precision, recall, and F1 on the test split. DDoS recall = 1.000 reflects zero missed attacks.

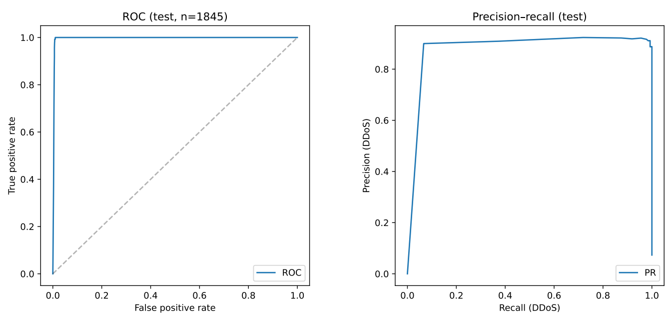
### E. Blind Test

To reduce evaluation bias from label order, `scripts/blind_test_random.py` draws a balanced random sample from the test split (model receives images only, no label information). With the maximum balanced sample of **270 images** (135 DDoS + 135 normal, seed 42), the model correctly classifies 267 of 270 images:

| True\Pred | DDoS | Normal |
|---|---|---|
| DDoS | 135 | 0 |
| Normal | 3 | 132 |
| Accuracy | **98.9%** | |
| Macro F1 | **0.9889** | |

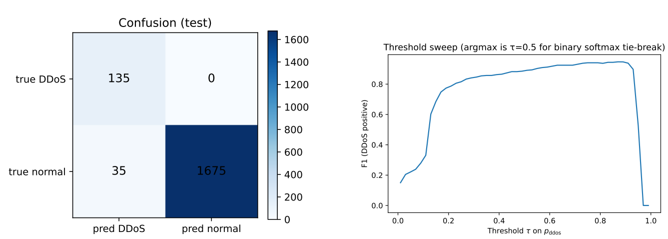
### *F. ROC, Precision–Recall, and Calibration*

Figure 4 shows the ROC and precision–recall curves on the test split (generated by `scripts/eval_roc_pr_calibration.py`). The calibration diagnostic yields a test **Expected Calibration Error (ECE)** $\approx$ **0.074** and a heuristic max-F1 threshold of $\approx 0.87$ on the DDoS softmax probability (Fig. 5).




(a) ROC curve (b) Precision–recall curve

Fig. 4. ROC and precision–recall curves for the primary 0.3 s model on the test split (via `run_all_paper_figures.sh`).




(a) Confusion matrix heatmap (b) Threshold vs. F1 sweep

Fig. 5. Test-split diagnostics: confusion heatmap (left) and F1 score as a function of the DDoS probability threshold (right). The heuristic max-F1 threshold is $\approx 0.87$.

### *G. Comparison with Baselines*

Table VI contrasts the primary model with a trivial majority classifier and the legacy per-packet configuration.

The majority-class baseline achieves 92.7% accuracy due to class imbalance, which confirms that accuracy alone is insufficient for security evaluation. The macro F1 of the primary model (0.9375) far exceeds the baseline (0.481), demonstrating meaningful attack detection beyond statistical luck.

TABLE VI
PERFORMANCE COMPARISON ACROSS CONFIGURATIONS ON RESPECTIVE TEST SETS.

| Configuration | $N$ | Acc. | Macro F1 |
|---|---|---|---|
| Always predict Normal | 1845 | 92.7% | 0.481 |
| Per-packet ckpt † on 0.3 s test (cross-domain) | 1845 | 89.9% | 0.473 |
| Per-packet ckpt † on per-packet test | 19 300 | 98.8% | 0.497 |
| Per-packet ckpt † blind test | 3000 | 99.9% | 0.9993 |
| **Ours (0.3 s, primary)** | **1845** | **98.10%** | **0.9375** |

† `model/resnext50_32x4d_best.ckpt`

## V. Discussion

### *A. Cross-Domain Representation Mismatch*

A critical finding is that the two image representations (per-packet vs. 0.3 s time-window) are **not interchangeable** without retraining. When the 0.3 s checkpoint is applied to per-packet test images, macro F1 drops to 0.497 with argmax never predicting DDoS (19 300 samples; 2026-04-03 run). Symmetrically, the per-packet checkpoint applied to the 0.3 s test set yields macro F1 of 0.473 with zero DDoS recall. This finding underlines a fundamental requirement: **deployment must use the same image construction as training**.

### *B. Operational Considerations*

The live inference service (`run_pipeline.py`) processes per-packet images within rotating 0.3 s PCAP windows. Each window logs a prediction $pred \in \{\text{ddos}, \text{normal}\}$ and the DDoS probability $p_{\text{ddos}}$. On public interfaces, Internet background traffic—scans, probing, heterogeneous protocols—may differ substantially from CICDDoS2019 training traffic, causing elevated false-positive rates. Consequently, $p_{\text{ddos}}$ values from the deployed model represent **classifier suspicion**, not confirmed incidents. Operational use cases should layer threshold tuning on $p_{\text{ddos}}$, consecutive-window voting, or hybrid rules before triggering automated responses.

### *C. Limitations*

- Evaluation is **same-domain** (train and test from CICDDoS2019); generalization to other datasets or enterprise traffic is not validated.
- The SYN-count labeling heuristic may mislabel PCAPs containing connection-heavy benign services.
- The system covers only TCP SYN floods; UDP amplification, HTTP floods, and application-layer attacks are out of scope.
- Adversarial crafting of packet bytes to evade the image classifier is not evaluated.

## VI. Conclusion and Future Work

We presented *SYNSheild*, an end-to-end deep learning pipeline for detecting SYN flood DDoS attacks by classifying 0.3 s time-window packet images with a ResNeXt-50 model trained on Huawei MindSpore. The primary model achieves **98.10%** test accuracy, **0.9375** macro F1, and **100% recall on the DDoS class** with zero missed attacks. A blind evaluation on 270 balanced samples confirms strong performance at 98.9% accuracy and macro F1 of 0.9889. Cross-domain experiments reveal that representation alignment between training and deployment is non-negotiable; operational deployments that violate this constraint exhibit near-random DDoS detection.

**Future work** includes: (1) validation on additional DDoS families (UDP, HTTP-flood) and out-of-distribution packet traces; (2) systematic comparison of window lengths (0.3 s, 1 s, 5 s) and dual-view fusion (SYN-only vs. all-packets); (3) hardware export to Huawei Ascend NPU for edge deployment; (4) calibrated alert strategies (thresholded $p_{\mathrm{ddos}}$, temporal smoothing) to reduce false positives on live Internet traffic; (5) adversarial robustness evaluation and domain adaptation under concept drift.

## ACKNOWLEDGMENT

The authors thank Eng. Mustafa Bizri (Beirut Arab University) for project supervision and guidance throughout this work. They also thank the Canadian Institute for Cybersecurity for releasing the CICDDoS2019 dataset, and the Huawei MindSpore and *mindcv* teams for the open-source deep learning framework used in all experiments.

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